

Side Brush & FlexiArm Gearboxes

OOM-M02-02 / OOM-M02-03 / Community Design Reference

DOCUMENT TYPE

COMMUNITY DESIGN REFERENCE

REVISION

A

DATE

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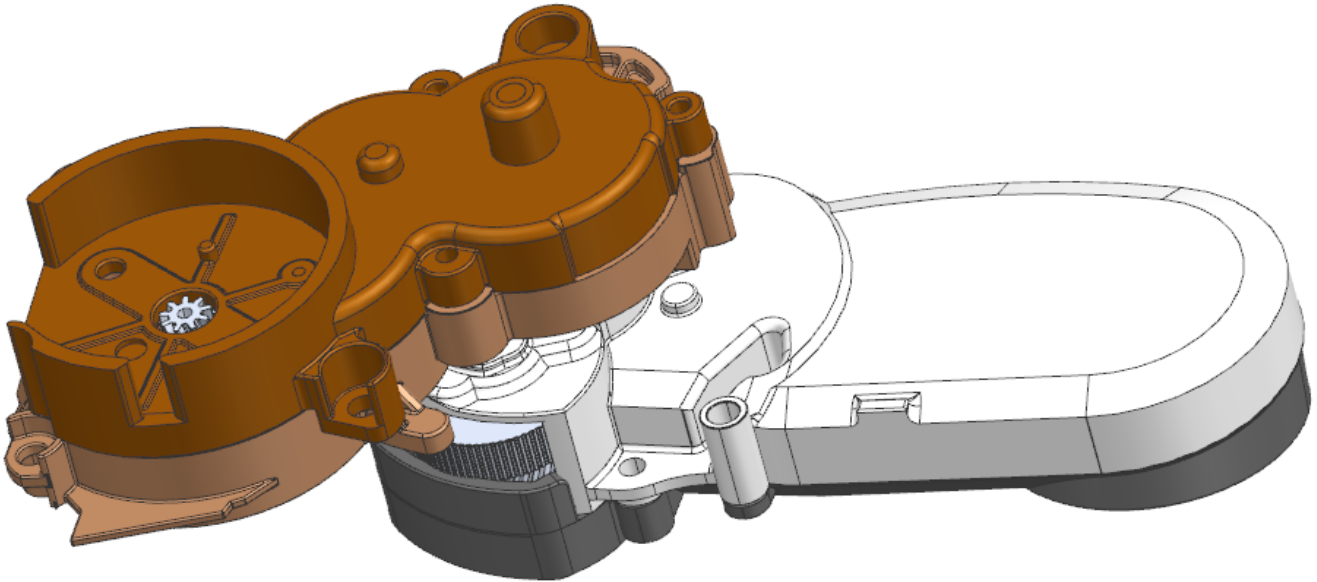


Figure 1. OOM-M02 side-brush module. The OOM-M02-03 brush gearbox and its motor (upper left, cover removed) are carried on the OOM-M02-02 FlexiArm extendable arm (right). The two housings meet at an angled joint.

Purpose. Explain the two gearboxes inside the OOM-M02 side-brush module for the OOMWOO One community: the OOM-M02-03 three-stage brush reduction (50.248:1), the OOM-M02-02 eleven-gear FlexiArm transfer train (39/14), their gear geometry, compound-axis logic, housing interfaces and the helical stages that carry torque through the arm joint.

COMMUNITY USE / Use this document together with the native CAD and STEP files. The tables explain the design intent and parameter relationships; the current repository CAD remains the master geometry for manufacturing, remixing or modification.

DATA STATUS / Revision A is built from two gear-parameter sheets, two motion studies and annotated CAD screenshots. Values are tagged **VERIFIED** when they reproduce from the source data, **DERIVED** when calculated in this document, and **CONFIRM** when they are an interpretation that must be checked against the CAD. Purchased bearing and bushing content is not identified in the source set and is left open in Section 7.

About This Design Reference

A practical engineering companion for understanding, rebuilding and modifying the two gearboxes of the OOM-M02 side-brush module.

OOMWOO One is an open-source robot vacuum project intended to be buildable, documented and hackable. The OOM-M02 module carries two independent gear trains with different jobs. OOM-M02-03 turns motor speed into brush speed at a high ratio; OOM-M02-02 routes rotation along the length of the extendable arm and through its angled joint at a ratio close to unity. This reference keeps the two separate so the ratio logic, the compound axes and the housing relationships stay easy to follow.

How to use this document

- **Part-code traceability:** custom components are identified by the project codes read from the CAD feature trees (Section 1.1). Codes that appear without a role label are listed as observed rather than guessed.
- **CAD-first reading:** tooth counts, modules, pressure and helix angles and centre distances explain the design; the repository CAD/STEP geometry remains the manufacturing master.
- **Kinematic clarity:** each gearbox has its own chain diagram, with compound (coaxial) members marked separately from external meshes so it is clear where torque changes shaft.
- **Explicit confidence:** the three cover tags appear throughout. Nothing that could not be read from the source set is presented as a dimension.
- **Data review:** Section 8 lists three internal inconsistencies found in the parameter sheets. Close those against the CAD before manufacturing or committing a remix.

Design basis

ID	Design basis	Defines	Use here
B1	OOM-M02 CAD / STEP set	Gear geometry, housing interfaces, shaft and support features	Manufacturing master
B2	Gear parameter sheet, 31 AUG 2026	OOM-M02-03 tooth counts, modules, pressure and helix angles, diameters, centre distances	Sections 2, 3
B3	Gear parameter sheet, 02 SEP 2026	OOM-M02-02 eleven-gear mesh list with diameters and centre distances	Sections 4, 5
B4	Motion study, 01 SEP 2026	OOM-M02-03 gear train running under a rotary motor driver	Compound-axis confirmation
B5	Motion study, 03 SEP 2026	Complete OOM-M02 module with the arm sectioned	Module architecture, joint geometry
B6	Annotated CAD screenshots	Feature-tree part codes, gear position labels Z1-Z11	Part numbering, position mapping
B7	OOM-M03-01 reference, Rev B	Series document conventions and addendum/root formulae	Cross-module consistency

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1 Module Architecture & Part Numbering

Two gearboxes, two housings and one angled joint: a high-ratio brush drive carried on a near-unity transfer train that reaches along the extendable arm.

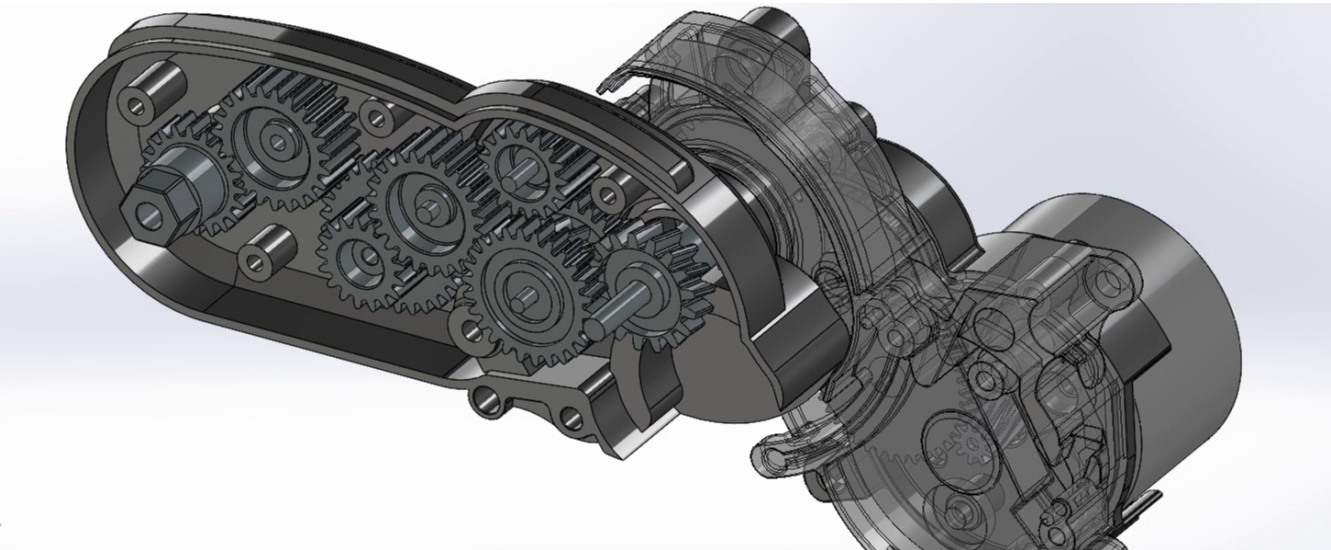


Figure 2. Complete OOM-M02 module from the 03 SEP motion study. The OOM-M02-02 arm is sectioned to expose its eleven-gear train; the OOM-M02-03 brush gearbox and motor are shown transparent on the right. The gear plane of the arm and the gear plane of the brush module are not parallel, and the helical group sits at that break.

Part code	Designation	Type	Primary role
OOM-M02-01	Module-level component	Custom part	Top-level member of the module assembly, outside both gear cavities. Role per CAD. CONFIRM
OOM-M02-02	FlexiArm extendable arm gearbox	Sub-assembly, rev RV02	Long oval two-shell housing carrying the eleven-gear transfer train Z1-Z11 and the external hex drive interface.
OOM-M02-03	Side-brush motor gearbox	Sub-assembly, rev rv02	Brush-side housing carrying the DC motor and the three-stage 50.248:1 reduction Z1-Z6.
OOM-M02-04	Module-level component	Custom part, rev RV01	Assembled together with OOM-M02-03 in the brush-side sub-assembly. Role per CAD. CONFIRM

TWO GEARBOXES, ONE MODULE / OOM-M02-03 is a reduction gearbox: three stages, 50.248:1, with the module rising along the train. OOM-M02-02 is a transfer gearbox: seven external meshes on the main branch that telescope to only 39/14 = 2.7857:1. Reading either one as if it were the other type leads to the wrong conclusions about tooth loading and backlash budget.

Why the arm train is nearly unity

Eleven gears producing less than 3:1 look wasteful until the package is taken into account. The arm has to deliver rotation from one end of a long, slim, swinging housing to the other, then continue through a joint where the two gear planes break at an angle. The intermediate gears exist to relocate the torque path around bosses, the pivot and the housing wall, not to reduce speed. The same pattern appears in OOM-M03-01, where four spur meshes telescope to exactly 1.000 and the whole 3.091:1 comes from the helical input stage alone.

The practical consequence is that the intermediate gears in OOM-M02-02 all see roughly the same torque band, so the weakest tooth anywhere in the chain sets the limit rather than the last one. In OOM-M02-03 the opposite is true: torque rises by a factor of 50 from Z1 to Z6, which is why the module increases from 0.46 mm to 0.60 mm along the train.

1.1 CAD Component Inventory

Part codes exactly as they appear in the feature trees, with the gear-position mapping that follows from the tooth counts and instance quantities.

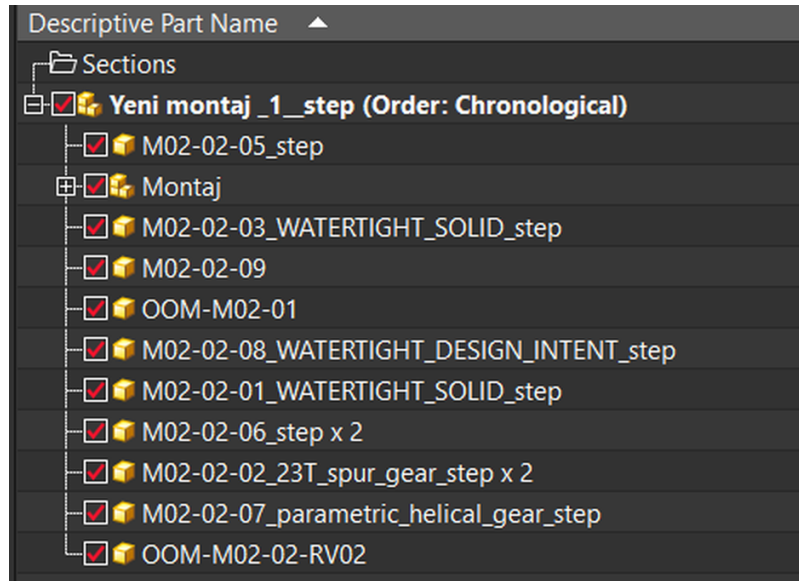


Figure 3. Component list of the OOM-M02 top-level assembly. Instance counts are significant: two 23T spur gear instances and two M02-02-06 instances match the repeated tooth counts in the parameter sheet.

OOM-M02-02 FlexiArm gearbox components

Code as modelled	Qty	Read as	Basis
M02-02-01_WATERTIGHT_SOLID	1	Housing shell	Watertight reconstructed solid. CONFIRM
M02-02-02_23T_spur_gear	2	Z2 and Z5, 23T m0.6	Only two 23T gears exist in the mesh list. DERIVED
M02-02-03_WATERTIGHT_SOLID	1	Housing shell	Second watertight solid. CONFIRM
M02-02-05	1	Per CAD	In the tree with no descriptive suffix. CONFIRM
M02-02-06	2	Z4 and Z6, 17T m0.6	Only two 17T gears exist in the mesh list. DERIVED
M02-02-07_parametric_helical_gear	1	17T helical, m_n 0.76	Z10 and Z11 are geometrically identical; one parametric body covers both. DERIVED
M02-02-08_WATERTIGHT_DESIGN_INTENT	1	Housing / structural solid	Design-intent rebuild, not a scan solid. CONFIRM
M02-02-09	1	Per CAD	In the tree with no descriptive suffix. CONFIRM

OOM-M02-03 brush gearbox components

Code as modelled	Qty	Read as	Basis
OOM-M02-03-01_design_intent	1	Housing, design intent	Gear-cavity shell rebuilt to design intent. CONFIRM
OOM-M02-03-02_SINGLE_SOLID	1	Housing / cover	Single-solid counterpart to the shell. CONFIRM
OOM-M02-03-03_1_motor	1	DC motor	Model identification not stated in the source set. CONFIRM
OOM-M02-03-03_2_helical_gear	1	Z1 motor pinion, 9T	The only helical pinion in the train; m_n 0.46, beta 18 deg. VERIFIED
0,5m z11	1	Z3, 11T m0.5	Name states module 0.5 and 11 teeth, matching Z3. VERIFIED

PART-CODE READING / The gear labels Z1-Z6 (brush gearbox) and Z1-Z11 (arm gearbox) restart independently in each parameter sheet. OOM-M02-02 Z1 is the 20T hex-drive spur gear; OOM-M02-03 Z1 is the 9T helical motor pinion. Always qualify a Z number with its module code.

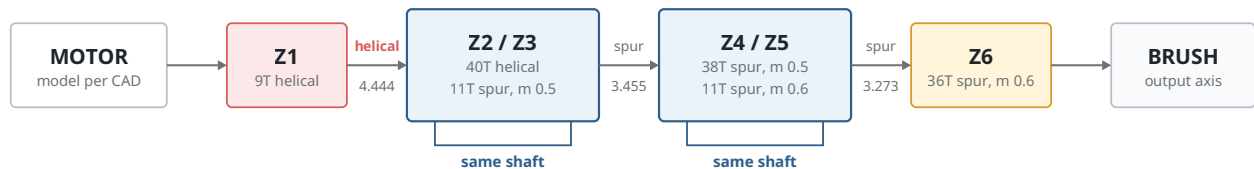
2 OOM-M02-03 Gear Train Design & Kinematics

A helical motor stage followed by two spur stages on two compound axes, with all three reductions stacking to 50.248:1 inside the brush-side housing.

50.248 : 1 OVERALL REDUCTION motor to brush output	1.990 % OUTPUT SPEED of motor speed	Opposite ROTATION motor vs. output
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OOM-M02-03 · KINEMATIC CHAIN

3 stages / 2 compound axes



$$(40/9) \times (38/11) \times (36/11) = 6080/121 = 50.248 : 1$$

3 external meshes (odd) → output reverses

Figure 4. OOM-M02-03 kinematic chain. Z2/Z3 and Z4/Z5 are compound members: each pair shares one axis, so all three ratios multiply.

Stage-by-stage ratio and centre distance

Stage	Mesh	Teeth	Module	Driven/driver	Centre distance	Function
Z1 -> Z2	Helical	9 -> 40	m _n 0.46	40/9 = 4.4444	11.850 mm	Primary reduction from the motor shaft
Z2 / Z3	Compound axis	40 / 11	0.46 / 0.5	1.000	Same axis	Rigid coaxial transfer, helical to spur
Z3 -> Z4	Spur	11 -> 38	m 0.5	38/11 = 3.4545	12.250 mm	Second reduction
Z4 / Z5	Compound axis	38 / 11	0.5 / 0.6	1.000	Same axis	Rigid coaxial transfer, module step-up
Z5 -> Z6	Spur	11 -> 36	m 0.6	36/11 = 3.2727	14.100 mm	Final reduction to the brush output

All three documented centre distances reproduce exactly from $a = (d_1 + d_2) / 2$ with no centre-distance modification. **VERIFIED**

Ratio arithmetic

$$i_{\text{total}} = (z_2/z_1) \times (z_4/z_3) \times (z_6/z_5) = (40/9) \times (38/11) \times (36/11) = 54720 / 1089 = 6080 / 121 = 50.2479$$

$$n_{\text{out}} / n_{\text{in}} = 121 / 6080 = 1.9901 \% \quad | \quad 3 \text{ external meshes (odd), so the output turns opposite to the motor.}$$

MODULE LADDER / The three stages use three different modules - 0.46, 0.50 and 0.60 mm - increasing in the direction of rising torque. When remixing, do not normalise them to one module: that would change every centre distance and the housing journal positions with them.

2.1 Compound Axes & Load Path

The 01 SEP motion study settles the question the parameter sheet cannot answer on its own: which gears share a shaft.

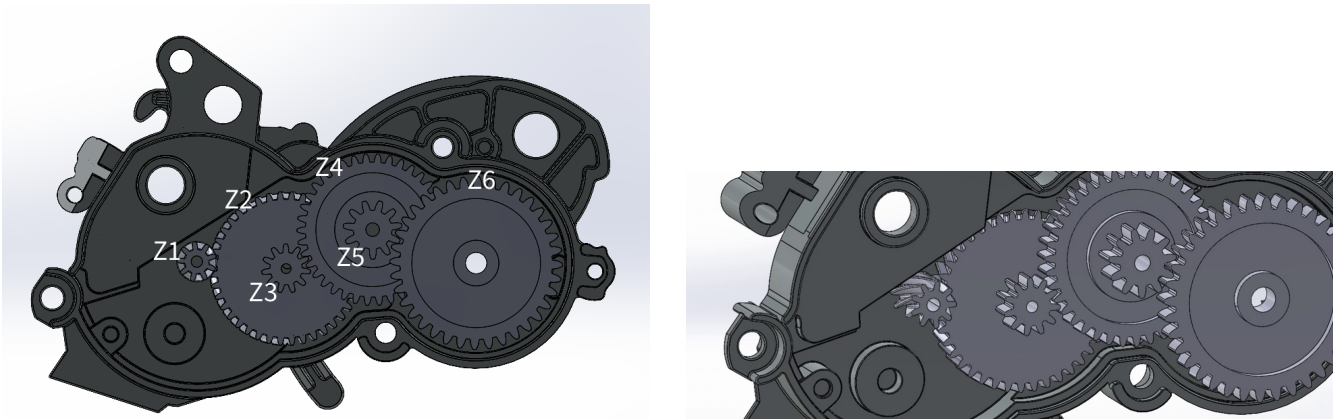


Figure 5. Left: OOM-M02-03 gear positions Z1-Z6 in the housing, running left to right. Right: frame from the motion study. The small spur pinions sit concentric with the large wheels behind them, which is the visual signature of the Z2/Z3 and Z4/Z5 compound members.

The parameter sheet lists three meshes but does not state how they connect. Three meshes among six gears can be wired as one chain with two compound axes, or as three unrelated pairs. Only the first reading produces a connected train, and the motion study confirms it directly: driving the motor axis turns all six gears, and each small pinion turns at exactly the speed of the large wheel behind it.

Axis	Members	Speed	Design consequence
Motor / Z1	Z1 9T helical pinion on the motor shaft	Input speed	Radial load plus helical axial thrust. Keep pinion overhang short and preserve motor axial retention.
Z2 / Z3	40T helical wheel + 11T m0.5 pinion	Input / 4.4444	First kinematic hinge. Helical thrust generated at Z1-Z2 is reacted here, so this axis needs a positive thrust path as well as radial support.
Z4 / Z5	38T m0.5 wheel + 11T m0.6 pinion	Input / 15.354	Pure spur loading: radial separating force only, but torque is already 15x motor torque.
Z6 / output	36T m0.6 output gear	Input / 50.248	Full output torque plus the external brush reaction. Highest-load support in the module.

Load-path summary

Motor torque enters at Z1, is multiplied 4.444 times at the helical mesh, passes coaxially from Z2 to Z3, is multiplied again at Z3-Z4 and Z5-Z6, and leaves through the Z6 axis to the brush. The housing absorbs the radial separating force at each of the three meshes, the helical thrust at the first stage only, and the external brush reaction at the output journal. Because torque rises monotonically, the Z6 journal and the Z4/Z5 axis are the two supports worth inspecting first after a run-in test.

THRUST IS LOCAL / Only the Z1-Z2 mesh is helical, so axial thrust exists on the motor axis and the Z2/Z3 axis and nowhere else. The two downstream spur stages need axial clearance to avoid face rub, not axial preload.

3 OOM-M02-03 Gear Geometry Reference

Documented tooth parameters, derived helical-system relationships and involute reference values for CAD recreation or cross-checking.

Helical input stage - Z1 / Z2

Gear	z	m _n	Pressure angle	Helix	Pitch d	Outside d _a	Root d _f	d _b at 25°	d _b at 20°
Z1 motor pinion	9	0.460	25 deg	18 deg	4.353	5.273	3.203	3.909	4.066
Z2 helical wheel	40	0.460	20 deg	18 deg	19.347	20.267	18.197	17.371	18.069

Spur transfer gears - Z3 through Z6

Gear	z	Module	Pressure angle	Pitch d	Outside d _a	Root d _f	d _b at 25°	d _b at 20°
Z3 on Z2 axis	11	0.500	25 deg	5.500	6.500	4.250	4.985	5.168
Z4	38	0.500	20 deg	19.000	20.000	17.750	17.220	17.854
Z5 on Z4 axis	11	0.600	25 deg	6.600	7.800	5.100	5.982	6.202
Z6 output	36	0.600	20 deg	21.600	22.800	20.100	19.576	20.297

Pitch, outside and root diameters are as documented and reproduce exactly **VERIFIED**. Base circles are calculated here; the parameter sheet lists base circle equal to pitch circle, which cannot be correct for a non-zero pressure angle (Section 8). **DERIVED**

HELICAL-SYSTEM INTERPRETATION / The documented pitch diameters satisfy $d = m_n z / \cos(\beta)$, so 0.460 mm behaves as a normal module. With $\beta = 18$ deg the derived transverse module is $m_t = 0.4837$ mm. Treating the pressure angle as normal gives a transverse pressure angle of 26.12 deg for a 25 deg normal angle, or 20.94 deg for a 20 deg normal angle. Preserve the native CAD helix hand and tooth form when remaking the pair.

Reading the diameter data

- For the spur stages the pitch diameters follow $d = m z$, the outside diameters follow $d_a = d + 2m$, and the documented root diameters follow $d_f = d - 2.5m$. This is the same tooth-proportion system used in OOM-M03-01, so the two modules share one convention.
- For the helical pair the same offsets are applied using the 0.46 mm normal module while the pitch diameter is enlarged by $1 / \cos(\beta) = 1.0515$.
- Both base-circle columns are given because the parameter sheet assigns 25 deg to each pinion and 20 deg to each wheel. A mesh must use one common pressure angle, so exactly one column applies to any given pair. Read the correct value from the CAD tooth profile and then use that column throughout.
- The 11-tooth pinions need no undercut relief at 25 deg. At 20 deg, 11 teeth sits below the classical 14-tooth undercut limit for a standard rack-generated profile, which is a further reason to settle the true pressure angle before remanufacturing.
- Two gears carry two different modules on one body: Z2/Z3 pairs m_n 0.46 with m 0.5, and Z4/Z5 pairs m 0.5 with m 0.6. Treat each compound as a single functional part even if it is modelled as two solids.

4 OOM-M02-02 Architecture & Mesh List

Eleven gears, eight meshes and nine axes inside the arm: a spur transfer path along the housing followed by a three-gear helical group at the joint.

2.7857 : 1 TELESCOPED RATIO 39/14, main branch	35.90 % OUTPUT SPEED of drive-axis speed	9 axes / 8 meshes TOPOLOGY two compound members
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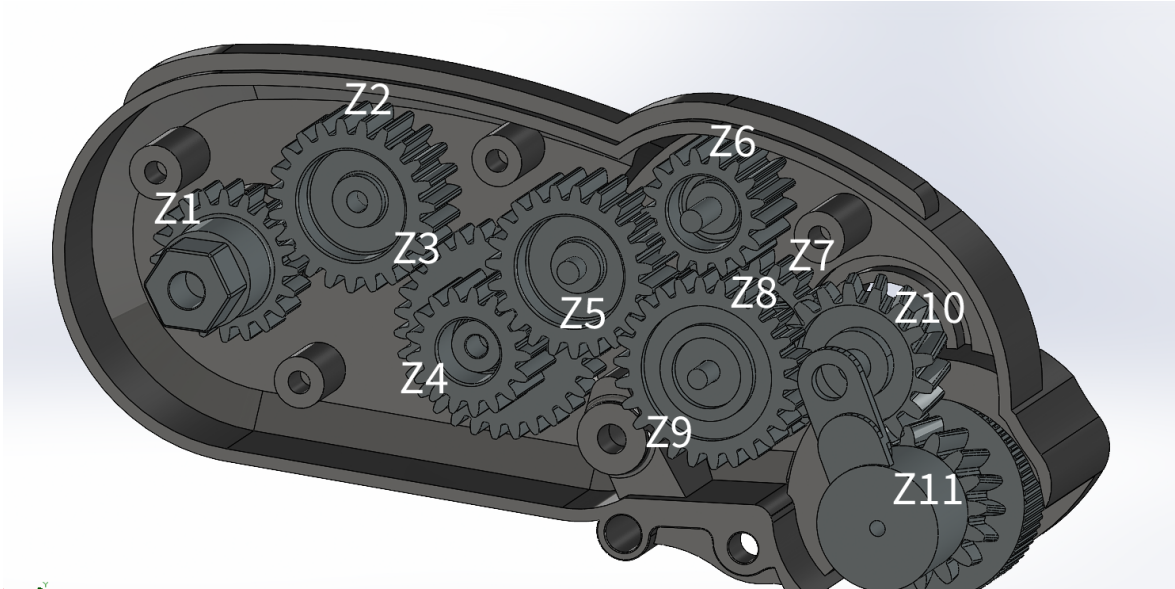


Figure 6. OOM-M02-02 gear positions Z1-Z11. Z1 carries the external hex form at the wide end of the arm; the Z8/Z10/Z11 helical group sits at the narrow end where the housing breaks toward the brush module.

Documented mesh list

Mesh	Type	Teeth	Ratio	a documented	a recomputed	Note
Z1 - Z2	Spur	20 - 23	1.1500	11.900 mm	12.900 mm	Mismatch of exactly 1.000 mm; Section 8. CONFIRM
Z2 - Z3	Spur	23 - 30	1.3043	15.900 mm	15.900 mm	Exact VERIFIED
Z4 - Z5	Spur	17 - 23	1.3529	12.000 mm	12.000 mm	Exact VERIFIED
Z5 - Z6	Spur	23 - 17	0.7391	12.000 mm	12.000 mm	Exact VERIFIED
Z6 - Z7	Spur	17 - 26	1.5294	12.900 mm	12.900 mm	Exact VERIFIED
Z5 - Z9	Spur	23 - 26	1.1304	14.700 mm	14.700 mm	Second branch off the Z5 axis VERIFIED
Z8 - Z10	Helical	14 - 17	1.2143	12.196 mm	12.196 mm	Exact VERIFIED
Z10 - Z11	Helical	17 - 17	1.0000	13.376 mm	13.376 mm	Unity pair, pure relocation VERIFIED

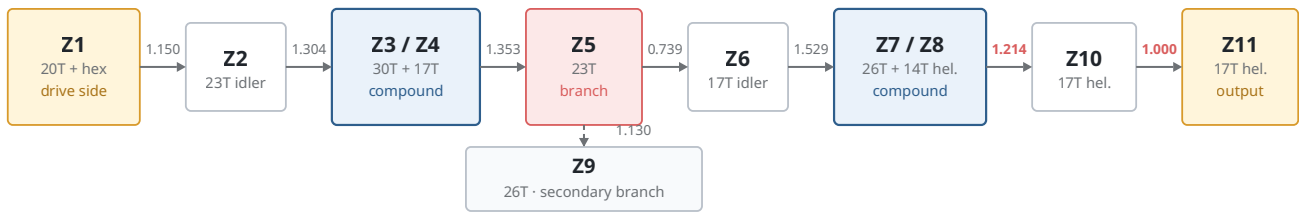
WHY THE COMPOUND AXES MATTER / Eight meshes among eleven gears leave the train disconnected unless two gear pairs share axes. Only one assignment makes the graph connected - Z3 with Z4, and Z7 with Z8 - and it is the same assignment the tooth counts and instance quantities suggest. Remove either compound and the arm becomes two or three unconnected gear groups in one housing, which the 03 SEP motion study rules out. These two assumptions carry Sections 4.1 and 5, so verify them first in the CAD.

4.1 Kinematic Chain & Telescoping Ratio

Seven external meshes on the main branch collapse to a single fraction, 39/14, because four of the eight tooth counts cancel in pairs across the Z5 axis.

OOM-M02-02 · KINEMATIC CHAIN

8 meshes / 9 axes / 2 compound axes



Main branch Z1 → Z11 = 39/14 = 2.7857 : 1

Secondary branch Z1 → Z9 = 39/17 = 2.2941 : 1

7 meshes (odd) → main output reverses
4 meshes (even) → Z9 keeps the drive sense

Figure 7. OOM-M02-02 kinematic chain. The chain is a branched tree: Z5 splits the path toward the joint group and toward Z9. Drive direction is discussed below.

$$i_{\text{main}} = (23/20)(30/23)(23/17)(17/23)(26/17)(17/14)(17/17) = (30/20) \times (26/14) = 39/14 = 2.7857$$

$$i_{\text{branch}} = (23/20)(30/23)(23/17)(26/23) = (30/20) \times (26/17) = 39/17 = 2.2941$$

Only three tooth counts survive: 20 and 30 on the first compound, and 26 with 14 on the second. Every other gear in the arm is a pure relocation member. Z2, Z5, Z6 and Z10 change nothing about the ratio; they carry torque past housing features and set where each axis lands. Z10-Z11 is a unity pair, so the last helical mesh exists purely to move the axis.

Axis map

Axis	Members	Meshes	Role
A1	Z1, 20T m0.6 + external hex form	1	External drive interface at the wide end of the arm.
A2	Z2, 23T m0.6	2	Idler; relocates the path past the first housing boss.
A3	Z3 30T + Z4 17T, compound	2	First compound member and the only source of the 30/20 factor. DERIVED
A4	Z5, 23T m0.6	3	Branch point: feeds Z6 toward the joint and Z9 on the secondary branch.
A5	Z6, 17T m0.6	2	Idler between the branch point and the joint group.
A6	Z7 26T spur + Z8 14T helical, compound	2	Second compound member and the spur-to-helical handover; source of the 26/14 factor. DERIVED
A7	Z10, 17T helical	2	Helical idler inside the joint region.
A8	Z11, 17T helical	1	End of the main branch; carries the link interface visible in the CAD. CONFIRM
A9	Z9, 26T m0.6	1	End of the secondary branch. Function not stated in the source set. CONFIRM

DRIVE-DIRECTION READING / The source set does not state which end of the arm is driven. This document presents the chain in the Z1 → Z11 direction, where the external hex form on Z1 acts as the input coupling and the train reduces 2.7857:1 toward the joint. If the CAD shows drive entering at the helical group instead, the identical chain reads as a 1 : 2.7857 step-up and the two interface roles swap. Every tooth count, centre distance and diameter in Sections 4 and 5 is unaffected either way; only the ratio direction and the torque grading change. Close this first, because it decides which tooth in the chain is the weakest. **CONFIRM**

5 OOM-M02-02 Gear Geometry Reference

One spur system at m 0.6 for the eight transfer gears, one helical system at m_n 0.76 and β 15 deg for the three joint gears.

Spur transfer gears - m 0.6

Gear	z	Module	Pressure angle	Pitch d	Outside d_a	Root d_f	d_b at 25°	d_b at 20°	Position
Z1	20	0.600	20 deg	12.000	13.200	10.500	10.876	11.276	Hex drive axis
Z2	23	0.600	20 deg	13.800	15.000	12.300	12.507	12.968	Idler
Z3	30	0.600	20 deg	18.000	19.200	16.500	16.314	16.915	Compound with Z4
Z4	17	0.600	25 deg	10.200	11.400	8.700	9.244	9.585	Compound with Z3
Z5	23	0.600	20 deg	13.800	15.000	12.300	12.507	12.968	Branch point
Z6	17	0.600	25 deg	10.200	11.400	8.700	9.244	9.585	Idler
Z7	26	0.600	20 deg	15.600	16.800	14.100	14.138	14.659	Compound with Z8
Z9	26	0.600	20 deg	15.600	16.800	14.100	14.138	14.659	Secondary branch

Helical joint gears - m_n 0.76, β 15 deg

Gear	z	m_n	Pressure angle	Helix	Pitch d	Outside d_a	Root d_f	d_b at 25°	Position
Z8	14	0.760	25 deg	15 deg	11.015	12.535	9.115	9.920	Compound with Z7
Z10	17	0.760	25 deg	15 deg	13.376	14.896	11.476	12.046	Helical idler
Z11	17	0.760	25 deg	15 deg	13.376	14.896	11.476	12.046	Main branch end

Geometry columns reproduce exactly from the parameter sheet, including the 25 / 20 deg pressure-angle split in the spur set **VERIFIED**. Base circles are calculated here **DERIVED**.

HELICAL-SYSTEM INTERPRETATION / The documented pitch diameters satisfy $d = m_n z / \cos(\beta)$, so 0.760 mm behaves as a normal module. With $\beta = 15$ deg the derived transverse module is $m_t = 0.7868$ mm and, treating 25 deg as the normal pressure angle, the transverse pressure angle is 25.77 deg. All three helical gears carry the same pressure angle in the source data, so this group needs no pressure-angle resolution - unlike the spur set.

Distinct-part summary

Distinct geometry	Qty	Positions
20T m0.6 spur + hex	1	Z1
23T m0.6 spur	2	Z2, Z5
30T m0.6 spur	1	Z3
17T m0.6 spur	2	Z4, Z6
26T m0.6 spur	2	Z7, Z9
14T helical m_n 0.76	1	Z8
17T helical m_n 0.76	2	Z10, Z11

Seven distinct geometries cover eleven installed gears, which matters when planning a print run or a spare set. Four of the eight spur gears are only two distinct parts.

Whether Z3+Z4 and Z7+Z8 are moulded as single compound bodies or stacked as two parts on a shared shaft is a CAD question. The instance counts in Figure 3 point to stacked separate parts: the two M02-02-06 instances account for Z4 and Z6, which leaves no spare 17T body for a one-piece Z3/Z4 compound. If they are separate parts, the rotational lock between them is a load-path element rather than a convenience. **CONFIRM**

6 Helical Stages & the Arm Joint

Both gearboxes use helical teeth, and in both cases the reason is geometric rather than acoustic.

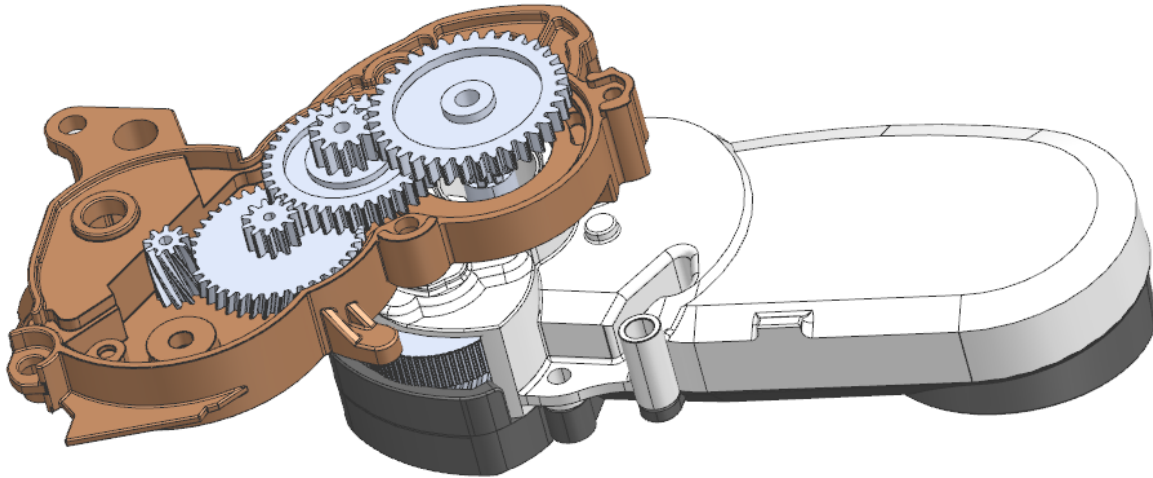


Figure 8. The brush gearbox opened on the arm. The brush-side gear plane and the arm-side gear plane are set at an angle to each other, and the helical group bridges that break.

The two helical systems compared

Property	OOM-M02-03, Z1 / Z2	OOM-M02-02, Z8 / Z10 / Z11
Normal module	0.460 mm	0.760 mm
Helix angle	18 deg	15 deg
Transverse module	0.4837 mm DERIVED	0.7868 mm DERIVED
Transverse pressure angle	26.12 deg at 25 deg normal, 20.94 deg at 20 deg normal	25.77 deg at 25 deg normal
Helical meshes	1	2
Ratio contribution	The full 4.4444 of the first stage	1.2143 at Z8-Z10, exactly 1.000 at Z10-Z11
Design reason	Quiet, high-ratio first stage directly on the motor shaft, where speed is highest and torque lowest	Carries rotation through the angled break between the arm and the brush module

Parallel or crossed axes?

Reading	Helix hands	Shaft angle	Consequence
Parallel-axis helical	Opposite hands, 15 deg each	0 deg	Conventional line contact, high load capacity, thrust reacted axially on each shaft. All three axes stay in one plane.
Crossed-axis helical	Same hand, 15 deg each	30 deg	Point contact, much lower load capacity and higher sliding velocity, but the drive turns through 30 deg without a bevel or worm. Matches the visible plane break in Figures 2 and 8.

Both readings satisfy $a = (d_1 + d_2) / 2$ with $d = m_n z / \cos(\beta)$, which is why the documented centre distances cannot distinguish them.

What does distinguish them is the helix hand of each gear and the relative orientation of the three axes in the CAD.

HAND AND SHAFT-ANGLE RULE / Read the helix hand of Z8, Z10 and Z11 directly from the native CAD before remaking any of them. If the group is crossed-axis at 30 deg the contact is point contact, and the face width, material pairing and lubrication decisions all change. A part that is dimensionally correct but handed wrong will mesh, run, and then fail early. **CONFIRM**

Thrust bookkeeping

- **OOM-M02-03:** thrust exists on the motor axis and on the Z2/Z3 compound axis. Both need a defined thrust reaction face; the two downstream spur axes need axial clearance only.
- **OOM-M02-02:** thrust exists on the Z7/Z8 compound axis and on the Z10 and Z11 axes. The Z7/Z8 member is critical because it carries a spur mesh and a helical mesh at the same time, so its support takes a radial separating force and an axial thrust simultaneously.
- In a crossed-axis reading the thrust direction on Z10 and Z11 changes relative to the parallel case, which is a second reason to close the hand question before detailing the supports.

7 Housing, Shaft & Support Architecture

What the source set establishes about the load path, and what it deliberately leaves open.

In OOM-M03-01 the support hardware was annotated in the CAD views, so that document could name an MR106ZZ ball bearing, a JSM-1416-05 polymer sleeve and an AMC30604 Oilite bush. The OOM-M02 source set carries no equivalent annotation. What the screenshots and motion studies do establish is the support concept at each axis: integral moulded posts and journal bosses in the housing shells, with a two-shell closure that traps the gear stack axially.

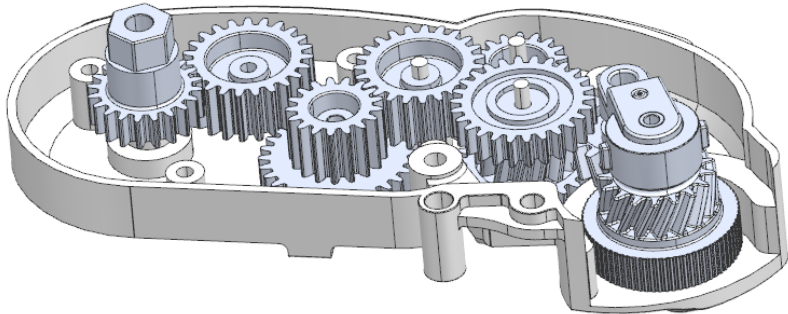


Figure 9. OOM-M02-02 gear set seated in the lower shell. The integral moulded posts are visible between the gears, the hex form sits on the Z1 axis at the left, and the helical group with its link interface is at the lower right, tilted out of the main gear plane.

Support concept by axis

Gearbox	Axis	Support concept observed	Load-path interpretation
M02-03	Motor / Z1	Motor internal bearings; pinion on the motor shaft	Radial load plus helical axial thrust. Keep pinion overhang short.
M02-03	Z2 / Z3	Fixed post or journal in the shells	First kinematic hinge; play here affects the motor mesh and the whole downstream path.
M02-03	Z4 / Z5	Fixed post or journal in the shells	Spur loading only, at roughly 15x motor torque.
M02-03	Z6 / output	Output journal through the housing to the brush interface	Full output torque plus the external brush reaction. Gear-to-coupling concentricity must survive housing closure.
M02-02	Z1 hex axis	Journal with an external non-circular form	Positive torque transfer rather than friction. Concentricity of gear and hex is the controlling requirement.
M02-02	Z2, Z5, Z6, Z9	Integral moulded posts in the lower shell	Idler and branch axes. Post diameter and perpendicularity set the centre distances.
M02-02	Z3 / Z4	Shared shaft or post carrying both gears	Ratio-bearing compound. If the gears are separate parts, the lock between them is a load-path element.
M02-02	Z7 / Z8	Shared shaft at the spur-to-helical handover	Simultaneous radial separating force and helical thrust. The most demanding support in the arm.
M02-02	Z10, Z11	Journals in the angled joint region; link interface on Z11	Helical thrust plus the external link reaction. Joint stiffness directly affects mesh contact.

Support concepts are read from the CAD screenshots and motion studies. No purchased bearing or bushing part numbers, shaft diameters, lengths or fits are stated in the source set. **CONFIRM**

Key geometric relationships

- Every centre distance is created by the housing post and journal locations. Print shrinkage or fit tuning must not move those axes independently of each other.
- Axial clearance is needed on every rotating gear so that closing the shells does not create face rub. Helical members additionally need a defined thrust reaction path in one direction.
- The compound axes are the kinematic hinges of their trains. Angular or radial play at a compound axis propagates both upstream and downstream, unlike play at an idler.
- Treat every support as one tolerance chain: shaft or hub diameter, housing seat, axial retention and running clearance have to stay compatible. Use the CAD for support lengths and axial stack-up.
- Both housings are two-shell designs whose perimeter register preserves gear-axis spacing after closure, and it should not distort an output journal or remove end clearance.

WHAT TO ADD IN REV B / The highest-value additions to this document are the purchased support hardware per axis, the shaft diameters and lengths, and the helix hands of Z8, Z10 and Z11. Those three items close nearly every open point in Sections 6 and 7.

8 Parameter-Sheet Consistency Review

Three internal inconsistencies were found while cross-checking the two parameter sheets. None of them changes the gear train architecture, and all three should be closed against the CAD before manufacture.

#	Finding	Where	Why it cannot be right as written	Recommended resolution
1	Pressure angle split 25 / 20 deg within a single mesh	Every documented pair in both sheets except the m_n 0.76 helical group	Two involute gears run conjugately only if they share one pressure angle in the plane of action. A 25 deg pinion cannot mesh correctly with a 20 deg wheel of the same module.	Read one pressure angle per mesh from the CAD tooth profile, then apply it to both members. Sections 3 and 5 give base circles for both readings, so either resolution is usable immediately.
2	Base circle listed equal to pitch circle	All diameter tables in both sheets: base, pitch and rolling circle share one value on every row	$d_b = d \cos(\alpha)$ is strictly smaller than the pitch circle for any non-zero pressure angle. Equality would mean a 0 deg pressure angle.	Treat the base-circle column in the source sheets as unpopulated. This document derives base circles at both 20 and 25 deg for every gear; use those, or regenerate from the CAD.
3	Z1-Z2 centre distance 11.900 mm	OOM-M02-02, first documented pair	For m 0.6 with 20T and 23T, $a = (12.000 + 13.800) / 2 = 12.900$ mm. No integer tooth count at m 0.6 gives 11.900 mm against a 23T wheel, and the neighbouring Z6-Z7 pair is correctly listed as 12.900 mm.	Read as a transcription error for 12.900 mm. Confirm against the CAD axis positions; if the CAD really shows 11.900 mm, then a tooth count or the module differs from the sheet.

What did cross-check exactly

- All three OOM-M02-03 centre distances (11.850, 12.250, 14.100 mm) reproduce exactly from the tooth counts and modules. **VERIFIED**
- Seven of the eight OOM-M02-02 centre distances reproduce exactly; only finding 3 above does not. **VERIFIED**
- Every outside and root diameter in both sheets reproduces exactly from $d_a = d + 2m$ and $d_f = d - 2.5m$, including the helical rows using the normal module. **VERIFIED**
- Both helical systems satisfy $d = m_n z / \cos(\beta)$ to three decimal places, confirming that the tabulated modules are normal modules rather than transverse. **VERIFIED**
- The OOM-M02-02 mesh list forms a connected tree under exactly one coaxial assignment, and that assignment is consistent with the instance counts in the CAD tree. **DERIVED**

PRIORITY / Finding 1 is the one to close first. It changes base circles, undercut margin on the 11-tooth pinions in OOM-M02-03, tooth thickness and therefore the backlash budget at every mesh. Findings 2 and 3 are bookkeeping by comparison.

9 Bill of Materials

Core parts of both gearboxes, with the items the source set does not identify marked rather than guessed.

OOM-M02-03 side-brush gearbox

#	Part / code	Description	Qty	Origin	Design detail
1	OOM-M02-03-01	Housing, design-intent shell	1	Custom CAD	Gear cavity and motor pocket
2	OOM-M02-03-02	Housing / cover, single solid	1	Custom CAD	Closes the cavity; carries upper supports
3	OOM-M02-03-03_2	Z1 motor pinion, 9T helical	1	Custom CAD	m_n 0.46, beta 18 deg; motor-shaft input
4	per CAD	Z2 / Z3 compound: 40T helical + 11T spur	1	Custom CAD	m_n 0.46 with m 0.5 on one axis
5	0,5m z11	11T m0.5 spur, Z3 geometry	1	Custom CAD	Named in the CAD tree
6	per CAD	Z4 / Z5 compound: 38T + 11T spur	1	Custom CAD	m 0.5 with m 0.6 on one axis
7	per CAD	Z6 output gear, 36T spur	1	Custom CAD	m 0.6; drives the brush interface
8	OOM-M02-03-03_1	DC motor	1	Purchased	Model per CAD/BOM CONFIRM
9	per CAD	Gear shafts / posts, bearings, bushings	per CAD	Per CAD/BOM	Not identified in the source set CONFIRM

OOM-M02-02 FlexiArm gearbox

#	Part / code	Description	Qty	Origin	Design detail
1	M02-02-01	Housing shell, watertight solid	1	Custom CAD	One half of the gear cavity
2	M02-02-03	Housing shell, watertight solid	1	Custom CAD	Second shell
3	M02-02-08	Housing / structural solid, design intent	1	Custom CAD	Design-intent rebuild
4	per CAD	Z1, 20T m0.6 spur with external hex form	1	Custom CAD	External drive interface
5	M02-02-02	23T m0.6 spur, Z2 and Z5	2	Custom CAD	Identical parts
6	per CAD	Z3, 30T m0.6 spur	1	Custom CAD	Compound with Z4; ratio-bearing
7	M02-02-06	17T m0.6 spur, Z4 and Z6	2	Custom CAD	Identical parts
8	per CAD	Z7, 26T m0.6 spur	1	Custom CAD	Compound with Z8
9	per CAD	Z9, 26T m0.6 spur	1	Custom CAD	Secondary branch end
10	per CAD	Z8, 14T helical m_n 0.76	1	Custom CAD	beta 15 deg; hand per CAD CONFIRM
11	M02-02-07	17T helical m_n 0.76, Z10 and Z11	2	Custom CAD	Parametric body; hand per CAD CONFIRM
12	M02-02-05 / -09	Components present in the CAD tree	1 each	Custom CAD	Role per CAD CONFIRM
13	per CAD	Gear shafts / posts, bearings, bushings, link	per CAD	Per CAD/BOM	Not identified in the source set CONFIRM

Compound members are listed as one line item each because they act as one functional part. Whether a compound is a single moulded body or two gears on a shared shaft is the subject of open item O5. **CONFIRM**

9.1 Assembly & Manufacturing Notes

A practical build sequence for community reproductions, and the parameters most easily lost in a CAD remix.

Suggested assembly sequence

1. Inspect all housing journal and post surfaces, gear bores, the motor pocket and the hex interface. Remove only intentional support material or debris from functional surfaces.
2. Fit the OOM-M02-03 Z1 helical pinion to the motor shaft at the CAD axial position, using a method that does not drive load through the motor bearings.
3. Seat the motor in the brush-side housing, then place the Z2/Z3 compound so the helical teeth enter mesh without forcing. Install the Z4/Z5 compound, then the Z6 output gear.
4. Build the arm gear set in the lower shell in axis order: Z1 with its hex form, Z2, the Z3/Z4 compound, Z5, Z6, the Z7/Z8 compound, then Z10 and Z11. Install Z9 last so the branch mesh can be checked on its own.
5. Verify helix hand and axial position on Z8, Z10 and Z11 before closing the joint region. A wrong hand will still assemble.
6. Close each housing lightly and rotate each train by hand through several revolutions. Expect motor cogging in the brush gearbox, but no hard interference, housing rub or tight spot in either train.
7. Tighten each housing evenly using the current CAD/BOM fastener strategy, repeat the rotation check, then begin powered run-in at reduced voltage before full-load use.

Manufacturing and modification notes

- Preserve the helix hand, helix angle, normal module and compound coaxial relationships when moving either gearbox between CAD systems. Helix hand is the parameter most often lost in a STEP round-trip through a rebuild.
- Do not normalise the three modules in OOM-M02-03 to a single value. The module ladder is deliberate and every centre distance depends on it.
- Do not assume the two 17T gears and the two 23T gears are interchangeable with the compound members. Check bore and hub features, which differ even where the tooth geometry does not.
- If journal fits are printed, tune them with process-specific test coupons rather than globally scaling the housing.
- Record material, print process, orientation, post-cure and dimensional compensation for gears and housings so another builder can reproduce the same behaviour.
- Use only a small amount of compatible gear lubricant if lubrication is required; keep it away from motor commutator openings and any plain-bearing material intended to run dry.

10 Community Build Notes

A compact inspection checklist for reproducing either gearbox, and the reference numbers to measure against.

What to look for after assembly

Check	What the builder should observe
Motor alignment	The OOM-M02-03 Z1 pinion should enter the Z2 helical mesh evenly across the intended face width, with no visible skew or end contact.
Helical mesh, brush side	Rotate slowly and confirm there is no hard spot. This stage carries both radial force and axial thrust.
Compound axes	Each compound member should run as one coaxial unit with no wobble between its two gears. Check Z2/Z3 and Z4/Z5 in the brush gearbox, Z3/Z4 and Z7/Z8 in the arm.
Brush output	The Z6 axis should remain concentric with the brush coupling and rotate without rubbing the housing.
Arm transfer path	Z2, Z5, Z6 and Z10 should rotate freely and preserve positive backlash around the full revolution. A tight spot at one angular position usually means a post is out of perpendicular.
Hex interface	The external hex should be concentric with the Z1 gear axis and transfer torque without deflecting the gear.
Joint group	Z8, Z10 and Z11 should mesh across full face width at the modelled shaft angle. Reduced contact at one end points to a hand or shaft-angle error, not a clearance problem.
Secondary branch	Z9 should rotate with Z5 with no added drag. Because it is a dead-end branch, extra friction there is easy to miss in an assembled test.
Housing closure	Fastening either set of shells should not significantly increase rotation torque or create gear-face rub.
Run-in	Start at reduced voltage, record sound and current, then inspect teeth, journals and housing witness marks.
Remix documentation	Record any changed motor, shaft, journal, gear, hex or link dimension so the modified version stays reproducible.

Design reference measurements

OOM-M02-03	Reference
Z1-Z2 centre distance	11.850 mm
Z3-Z4 centre distance	12.250 mm
Z5-Z6 centre distance	14.100 mm
Stage ratios	4.4444 / 3.4545 / 3.2727
Overall ideal ratio	50.2479 : 1
Output speed	1.9901 % of motor
Output sense	Opposite to motor

OOM-M02-02	Reference
Z1-Z2 centre distance	12.900 mm, Section 8
Z2-Z3 / Z4-Z5 / Z5-Z6	15.900 / 12.000 / 12.000
Z6-Z7 / Z5-Z9	12.900 / 14.700 mm
Z8-Z10 / Z10-Z11	12.196 / 13.376 mm
Main branch ratio	39/14 = 2.7857 : 1
Secondary branch ratio	39/17 = 2.2941 : 1
Main output sense	Opposite to drive axis

MEASURE BEFORE ASSUMING / The reference values above are ideal geometry from the parameter sheets, not measured parts. Treat them as the target a printed or machined reproduction should be checked against, and note that the OOM-M02-02 Z1-Z2 value shown is the corrected 12.900 mm rather than the sheet value.

10.1 Open Items & References

What this revision could not close from the available source set, and where each answer lives.

Open items for Revision B

#	Open item	Decides	Where to read it
01	Drive direction of OOM-M02-02: hex axis or helical group	Ratio direction, torque grading, weakest tooth	Motor or coupling location in the module CAD; motion-study driver assignment
02	Helix hand of Z8, Z10, Z11 and the shaft angle of the joint group	Parallel vs crossed axis, contact type, load capacity	Native CAD gear features and axis orientations
03	One pressure angle per mesh	Base circles, tooth thickness, undercut margin, backlash	CAD tooth profile measurement
04	Function of the Z9 secondary branch	Whether it is a second output or a sensing / indexing path	What the Z9 axis connects to outside the gear cavity
05	Whether the compound members are single bodies or stacked parts	Torque-lock detailing between the two gears	Part structure of the Z3/Z4 and Z7/Z8 axes in CAD
06	Purchased support hardware, shaft diameters and lengths	Section 7 tolerance chains	Current CAD/BOM
07	Motor identification for OOM-M02-03	Input speed, torque and thrust budget; achievable brush speed	CAD/BOM motor component
08	Roles of OOM-M02-01, OOM-M02-04, M02-02-05 and M02-02-09	Module-level architecture completeness	Top-level assembly in CAD

References and project links

- [1] OOMWOO One CAD repository - <https://github.com/makerspet/oomwoo-one-cad>
- [2] OOMWOO One CAD 3D design specification - <https://github.com/makerspet/oomwoo-one-cad/blob/main/docs/SPEC.md>
- [3] OOMWOO main project BOM and design documentation - <https://github.com/makerspet/oomwoo>
- [4] OOM-M02-03 gear parameter sheet, 31 AUG 2026 - tooth counts, modules, pressure and helix angles, diameters, centre distances.
- [5] OOM-M02-02 gear parameter sheet, 02 SEP 2026 - eleven-gear mesh list with diameters and centre distances.
- [6] OOM-M02-03 motion study, 01 SEP 2026 - rotary-motor driven gear train, used for compound-axis confirmation.
- [7] OOM-M02 module motion study, 03 SEP 2026 - complete module with the arm sectioned, used for joint geometry.
- [8] Annotated CAD screenshots, SEP 2026 - feature-tree part codes and gear position labels Z1-Z11.
- [9] OOM-M03-01 Motor Drive Gearbox, Community Design Reference, Rev B, 28 AUG 2026 - series conventions and tooth-proportion formulae.

REPOSITORY NOTE / For current licence terms, contribution guidance, part revisions and manufacturing geometry, use the repository README, LICENSE and native CAD files. This PDF is an explanatory design reference, not a substitute for the current CAD revision.